Legal Document Interactive Application

presented by

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I. Introduction

This project was brought to me within my internship at Planck AI. This project aims at creating a streamlit application that aids in understanding of legal documents, making easy summarization for lawyers who may want to sift through documents quickly.

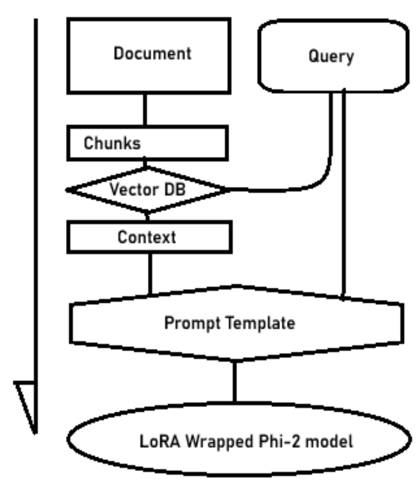
Our team was tasked with building training and deploying a few machine learning model techniques. In order to create further understanding we wanted to display: a summarization, identify each clause by type, and provide an interactive chatbot allowing the user to ask questions.

Models

Building the Summarization model we used Google's **T5 -base model**. T5 stands for Text-to-Text Transfer Transformer, this model is an encoder -decoder model that transfers one text into another form. These models are often used for tasks like translation from one language to another, these models are also used for summarization.

For building a clause classifier we decided to use a **logistic regression model** for simplicity and improved interoperability. Logistic regression models are often used for classification, the biggest reason for using this model is simplicity and interpretability as compared to our black box models.

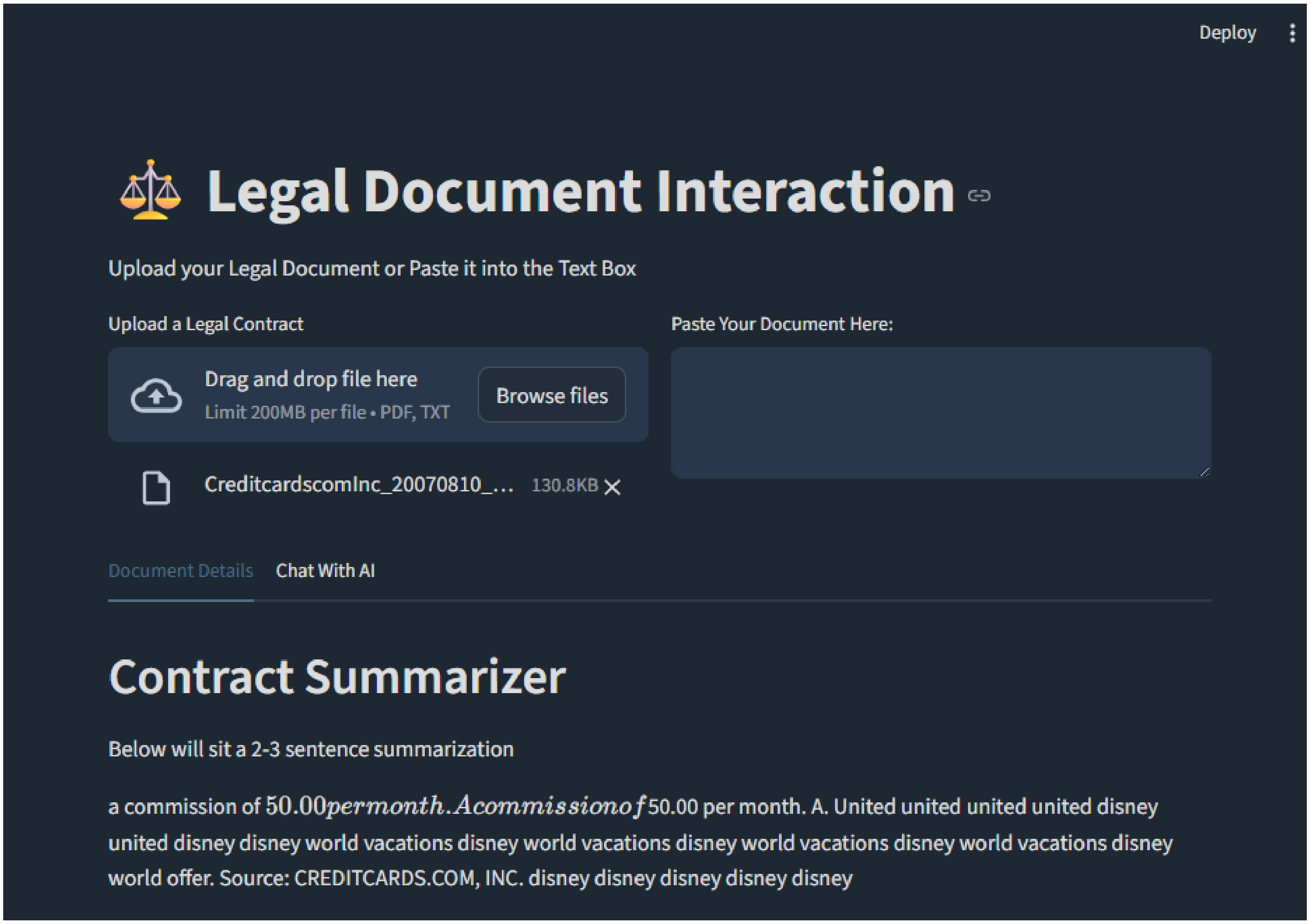
Our simple chat bot was created using multiple techniques. We used **Phi-2** as the base LLM, Phi- 2 is a smaller model initially created by Microsoft, having 2B parameters allowing for improved local performance. We also decided that we needed to finetune our model, to prevent re-writing Phi-2's weights we decided to use a **LoRA (Low Rank Adaption)** model. LoRA is used for finetuning models without re-writing the base LLM's weights. This can be important as it is often used for specific domains or tasks where you would want your model to be permanently stuck in.



RAG (Retrieval Augmented Generation) is used to give your model context, allowing the model to access your document. A RAG pipeline works by separating the document into chunks, embedding the chunks, and adding those embedded chunks into a vector database. When you ask the chat a question, the pipeline queries the vector database for chunks of text that are similar to the question, then gives these bits to the model as context. This prevents the model from having to look at the whole document and solves some context limitations of models.

II. Evaluation

T5 Summarization



As you can see above we have a summarization of the entered legal document. A few things to note about the quality. The summarization gives a somewhat coherent first sentence followed by a string of two or three words repeating. This quality is poor and leaves a lot to be desired. This model was not finetuned at all before implementing into our app due to time restrictions.

TF-IDF + Logistic Regression

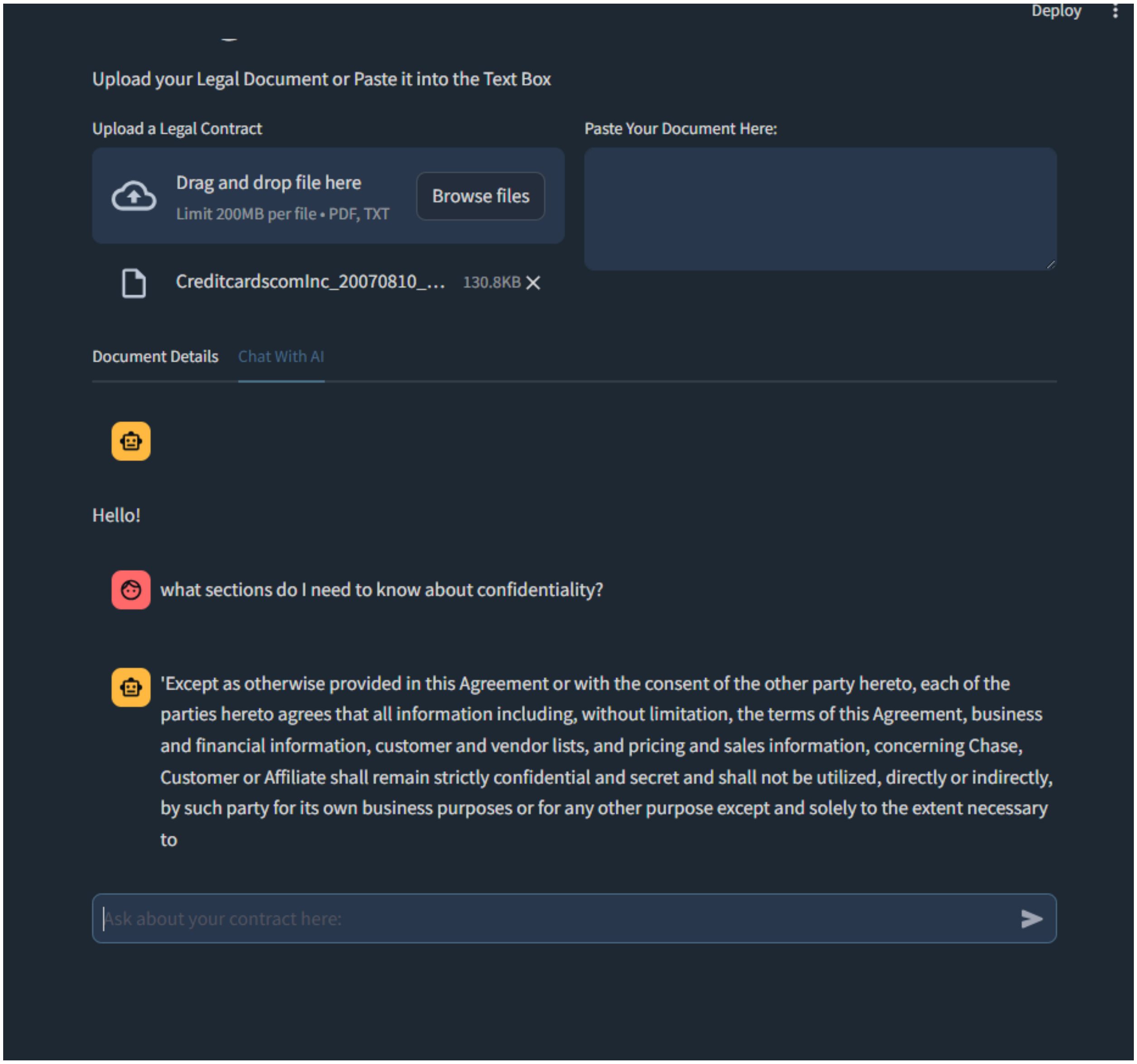
Contract Clause Individual Information

Below will sit a table containing the different clause types and the details that will contain them

	prob	pred	text
4	0.1351	Termination for Convenience	4. Approved Account For purposes of determining Affiliate's Commission, an "Approv
3	0.1392	Revenue-Profit Sharing	3. Referral Fee For each Approved Account (as defined in section 4 below) received th
7	0.0798	Revenue-Profit Sharing	8. Tracking of Sales Chase will be solely responsible for tracking Approved Accounts
9	0.0631	Post-termination Services	10. Chase Customers Customers who apply for Chase credit cards through the Chase
5	0.0573	Post-termination Services	6. Links Affiliate agrees to place Chase's links provided by Linkshare NetworkTM whi
21	0.0758	Non-Compete	22. Independent Investigation AFFILIATE ACKNOWLEDGES THAT IT HAS READ THIS A
22	0.1162	Minimum Commitment	23. Governing Law This Agreement will be governed in all respects by the laws of the
10	0.0852	Minimum Commitment	11. Product Descriptions Affiliate will only use credit card descriptions provided or a
6	0.08	Minimum Commitment	7. Order Processing Chase will be solely responsible for processing each order placed
8	0.0744	Minimum Commitment	9. Terms and Conditions of Credit Cards Chase is solely responsible for determining t

The image above is an example of a logistic regression model paired with TF-IDF. TF-IDF stands for Term Frequency - Inverse Document Frequency. This metric gives a value to each term in the sentence or paragraph and compares it to the paragraph as well as the rest of the document. This metric gives each term a value to indicate the uniqueness or importance of the term. We used our TF-IDF metrics to train a logistic regression model to classify the different types of legal clauses. As you can see in the display above we have the terms separated into sections by predicted clause type as well as displaying the logistic regressions model probability of the clause being assigned to the text snippet.

LoRA Finetuned Phi-2 RAG Chatbot



Our LoRA model was trained to return highlighted sections from the text to the user. After finetuning using the LoRA model it works rather well, however sometimes the format appears to be in poor quality, as you can tell the sentence returned by the assistant is incomplete.

Our LoRA model was trained on a RAG template that included the Context returned from querying the vector database, the initial query and providing the expected answer. This way we could get a better response throughout the whole RAG pipeline. The context was large and nearing the limitations of our LLM preventing us from including conversational context to our model, preventing follow up questions.

III. Course of Action

T5 Summarization

The T5 summarization section of our app is clearly lacking in quality. I believe we need to train a LoRA model on top of the T5 model to summarize our legal documents, leading to improved document summarization for this industry application. To do this we would need to go through each document and personally summarize the documents.

TF-IDF + Logistic Regression

Our TF-IDF model could be improved by editing the classification threshold of the model. We don't need to classify different sections if the probability of the clause is below 50%, I also believe that this would aid in cleanliness of the model. Applying a fine-tuned T5 model to summarize the text within each classification could prove beneficial at improving understanding.

LoRA Finetuned Phi-2 RAG Chatbot

Our current model is only fine-tuned to reproduce sections highlighted about questions you ask it. For our next steps I would like to create another LoRA model for more interactive questions and possibly advice, this would take more comprehensive data and possibly an improved base model. Experimenting with different base models could improve our output. I think creating an agentic structure to switch between different LoRA models could prove to be beneficial.

Application Interface

Noticing our chat model returns highlighted sections from our document, allowing the user to see that document and having the document automatically highlighted when asking different questions would be a cool display.

IV. Final Notes

This project gave me the opportunity to work through the full pipeline of building multiple machine learning models and deploying the models to an application accessible to users. Throughout my work on this project I have had opportunities to:

- **Managed end-to-end AI model development**, from data extraction and cleaning to fine-tuning LoRA and RAG pipelines, ensuring consistent progress.
- **Developed and fine-tuned transformer-based models** (Phi-2, T5) using **LoRA and RAG pipelines** to enhance legal document comprehension and question-answering performance.

- **Engineered an ETL data pipeline** to parse, clean, and format **legal contract data** (CUAD dataset) into JSONL for model training and evaluation.
- **Integrated TF-IDF and Logistic Regression** models into a **Streamlit web app**, enabling interactive document classification, clause tagging, and analytics visualization.
- **Designed and optimized a LangChain RAG pipeline**, improving document retrieval efficiency and model response accuracy through customized prompt templates and vector stores.
- **Researched transformer architectures and model evaluation techniques**, implementing insights into fine-tuning strategies and hyperparameter optimization for improved model performance.
- **Collaborated in weekly cross-functional meetings**, proposing benchmarking strategies and leading discussions on data labeling formats, LoRA integration, and evaluation benchmarks.